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LIPID AND BIOCHEMICAL CHANGES AFTER LOW-DOSE ORAL CONTRACEPTION

Diana F.M. Loke*, Charles S.A. Ng**,
Susan Holck***, Peter E Hall*** Shan S. Ratnam*

* Department of Obstetrics and Gynecology
National University of Singapore
National University Hospital, Singapore

** Division of Surgery
Department of Obstetrics and Gynecology
Singapore General Hospital, Singapore

*** World Health Organization
1211 Geneva 27, Switzerland

Abstract

A randomized double-blind study of the metabolic effects of 2 low-dose combined oral contraceptives was carried out in Singaporean women. The subjects comprised 58 women randomly allocated to two treatment groups (29 each): norethisterone 1 mg/ethinyl estradiol 35 μg (NET/EE) or levonorgestrel 150 μg/ethinyl estradiol 30 μg (LNG/EE) and a control group of 23 women using intra-uterine devices (IUD). Blood samples were taken on admission and at 3 and 12 months after pills or insertion of IUDs.

Fasting glucose levels and 2h glucose were decreased while triglyceride was increased throughout the treatment period in NET/EE group. LNG/EE group only showed significant suppression of 2h glucose at 12 months and LDL cholesterol at 3 months while total cholesterol was significantly suppressed at 3 and 12 months. The atherogenic index, LDL/HDL cholesterol was significantly reduced by 12 months. Both groups had no change in hemoglobin, hematocrit and total protein levels but alkaline phosphatase, bilirubin and aspartate transaminase (SGOT) were suppressed. While NET/EE suppressed albumin significantly, this was not observed with LNG/EE group. However, these differences observed with use of each pill preparations, were not so obvious between treatment groups and control. Changes in total, HDL and LDL cholesterol and SGOT were not significantly different than the IUD group. Furthermore, except for 2h glucose, there was no increase in the number of abnormal parameters after treatment. On the contrary, there was a reduction of abnormal values in most liver function parameters. Thus, except for glucose intolerance, the observed changes in metabolic parameters may not constitute any clinical significance. Differences between groups may be attributed to the progestogen content of the pill.

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Introduction

Oral contraception is a popular form of fertility control. Over the years with advances in research, the dosage of oral contraceptive pills has decreased markedly to reduce undesirable side effects. Many of these effects have been well documented, but most of these studies were carried out in the Western countries with few studies in the developing countries (1-5).

As a World Health Organization (WHO) Collaborating Centre in Research and Human Reproduction, Singapore participated in a multicentre study to compare the metabolic effects associated with the use of two current low-dose oral contraceptive preparations under the WHO Special Programme of Research, Development and Research Training in Human Reproduction. The pill preparations used were ethinyl estradiol with norethisterone or levonorgestrel. This report presents the lipid and biochemical (carbohydrate and liver function) changes in Singaporean women who had taken these preparations after 3 and 12 months.

Materials and Methods

Contraceptive pills

The following formulations were used:

a) 35 μg ethinyl estradiol with 1000 μg norethisterone (NET/EE group)
b) 30 μg ethinyl estradiol with 150 μg levonorgestrel (LNG/EE group)

Subjects

Subjects were recruited from the outpatient clinics and the Fertility Control Clinic of the University Department of Obstetrics and Gynecology, Kandang Kerbau Hospital. They were screened according to eligibility criteria set up for the study. The subjects were healthy female volunteers between ages of 18 and 34 with no known or suspected diabetes, liver diseases or thromboembolic disorders. At time of admission, subjects were at least 70 days post-partum. Fifty-eight subjects were recruited and randomly allocated to one of the 2 pill groups. Another 23 subjects, intending to use IUD were recruited as controls.

Blood samples were taken on admission, and 3 and 12 months after pills or insertion of IUDs. Venous blood was collected between day 15-26 of the menstrual cycle after an overnight fast of 12 hours. Each subject also received a physical examination at each of these visits.

Laboratory procedures

Measurements of serum glucose (both fasting and 2 h after glucose loading), triglycerides, cholesterol, HDL cholesterol, total protein, albumin, aspartate transaminase (SGOT), bilirubin and alkaline phosphatase were performed on each blood sample, by

automated methods on the Technicon Multianalyzer System (6-13). HDL cholesterol was precipitated by manganese phosphotungstate before estimation on the analyzer (14). LDL cholesterol was calculated using Friedewald's equation (15). Internal and external quality control schemes were established (16) and operated for this programme by the WHO Collaborating Centre for Research and Reference Service in Clinical Chemistry in Birmingham, United Kingdom.

Statistical methods

Within each study group, differences between admission and the 3- and 12-month follow-up visits were assessed by paired t-test. Differences in admission data between study groups were assessed by one-way analysis of variance. Significance of differences of changes (in % of admission value) at 3 and 12 months between study groups were assessed by one-way analysis of variance. The SAS statistical package was used for the analysis.

Results

All groups were matched for age, height, weight (or body mass index (BMI)), parity, SBP and DBP at admission (Table I). Except for SGOT, there were also no significant differences between groups for all measured parameters at admission.

Table I. Physical characteristics (mean±SD) of Singaporean women on low-dose oral contraceptives and IUD

Physical Characteristics	Groups		
	NET/EE	LNG/EE	IUD
Age (yrs)	25.4±4.2	26.6±3.1	26.0±4.0
Height (m)	1.55±0.06	1.56±0.07	1.58±0.07
Weight (kg)	51.2±7.2	52.9±8.8	56.3±7.3
BMI (kg/sq m)	21.4±3.1	21.7±3.4	22.5±2.6
Parity	1.3±0.8	1.4±0.5	1.5±0.7
SBP (mm Hg)	108.6±9.7	110.4±10.2	108.4±9.3
DBP (mm Hg)	73.2±6.8	75.1±8.0	72.6±7.3
No. of subjects	29	29	23

Table II. Glucose and lipid values (mean±SEM) at admission, 3 and 12 months after treatment in Singaporean women on low-dose oral contraceptives and IUD

	Fasting glucose (mg/dl)			Glucose 2h (mg/dl)			Triglycerides (mg/dl)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Admission	87.0±1.3	84.2±1.7	81.5±2.2	93.8±3.3	92.1±3.6	90.4±3.2	98.4±9.2	107.1±9.3	87.7±8.8
3 months	77.3±1.5***	82.1±1.8	82.1±1.9	107.0±4.0**	101.0±4.3	96.4±5.1	134.6±9.4***	101.1±8.3	82.4±9.0
12 months	79.7±1.7**	81.0±1.7	84.6±2.1	112.1±3.8***	105.6±4.3**	99.6±4.2*	119.7±8.3**	101.1±7.0	95.9±12.2

	Total cholesterol (mg/dl)			HDL cholesterol (mg/dl)			LDL cholesterol (mg/dl)		
Admission	196.9±6.2	207.4±7.1	198.1±7.4	43.6±1.4	40.8±2.0	42.9±2.4	134.2±6.7	144.3±6.6	145.2±6.3
3 months	188.3±6.0	195.0±7.1**	187.3±6.0	44.2±1.5	39.5±1.5	43.5±1.8	118.3±6.3***	133.7±7.5*	131.3±6.6
12 months	194.0±5.3	197.4±7.5*	193.3±7.5	47.6±2.2*	43.0±1.6	43.7±1.7	123.7±6.1*	133.9±7.3	135.3±8.5

	LDL/HDL cholesterol		
Admission	3.19±0.21	3.79±0.28	3.52±0.21
3 months	2.78±0.19**	3.57±0.32	3.12±0.21
12 months	2.80±0.24*	3.23±0.22*	3.18±0.24

* $p < 0.05$ vs corresponding admission value; ** $p < 0.01$ vs corresponding admission value; *** $p < 0.001$ vs corresponding admission value.

Within-group differences

Significant differences in mean fasting glucose and 2h glucose were observed in women on NET/EE at 3 and 12 months and in 2h glucose in women on LNG/EE or IUD at 12 months (Table II). Fasting glucose was suppressed while 2h glucose was raised after treatment. Mean triglyceride levels were significantly raised at both 3 and 12 months in women on NET/EE while they remained unchanged in women on LNG/EE or IUD. Total cholesterol was significantly reduced at 3 and 12 months in women on LNG/EE while it remained unchanged in other groups. Women on NET/EE had raised HDL cholesterol at 12 months and decreased LDL cholesterol at 3 and 12 months. While mean levels of LDL cholesterol in both women on NET/EE or IUD were decreased, they were not significantly different from admission values except for the level at 3 months in women on LNG/EE. The atherogenic index, LDL/HDL cholesterol ratio, was significantly reduced in women on NET/EE at 3 and 12 months and in women on LNG/EE at 12 months only.

Total protein, hemoglobin and hematocrit remained unchanged throughout the treatment period in all groups (Table III). In women on NET/EE, albumin and alkaline phosphatase were significantly decreased at both 3 and 12 months while total bilirubin and SGOT were significantly decreased at 3 months, returning to baseline at 12 months. Albumin remained unchanged for women on LNG/EE or IUD. Mean levels of total bilirubin, alkaline phosphatase and SGOT were lower after treatment in women on LNG/EE. However, levels of total bilirubin at 12 months and SGOT at 6 months were not significantly different from the corresponding admission value. For the IUD group, mean levels of alkaline phosphatase at 12 months and SGOT at 3 and 12 months were significantly lower than the corresponding admission value.

Between-group differences

No significant differences between groups were observed in 2h glucose, hemoglobin and hematocrit levels after treatment (Fig. 1). Fasting glucose of women on NET/EE at 3 and 12 months was significantly lower than the corresponding levels of women on LNG/EE and IUD at 3 months and that of women on IUD at 12 months, respectively. Minimal differences between groups were observed in lipid parameters except for triglycerides (Fig. 2). Women on NET/EE had significantly higher triglyceride levels at 3 months than women on LNG/EE or IUD. No significant differences were observed between groups for total protein and SGOT (Fig. 3). NET/EE induced significantly lower albumin and total bilirubin at 3 months and alkaline phosphatase at 3 and 12 months than LNG/EE or IUD.

Table III. Total protein, albumin, total bilirubin, alkaline phosphatase, aspartate transaminase, hemoglobin and hematocrit values (mean±SEM) at admission, 3 and 12 months after treatment in Singaporean women on low-dose oral contraceptives and IUD

	Total protein (g/dl)			Albumin (g/dl)			Total bilirubin (mg/dl)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Admission	7.77±0.06	7.70±0.08	7.68±0.07	4.62±0.03	4.54±0.04	4.54±0.05	0.83±0.06	0.84±0.08	0.84±0.06
3 months	7.79±0.07	7.66±0.06	7.57±0.10	4.38±0.04***	4.47±0.04	4.49±0.05	0.67±0.06***	0.69±0.05**	0.85±0.06
12 months	7.77±0.07	7.60±0.08	7.53±0.10	4.49±0.03**	4.53±0.04	4.46±0.07	0.74±0.05	0.77±0.05	0.82±0.09

	Alkaline phosphatase (U/l)			Aspartate transaminase (U/l)			Hemoglobin (g/dl)		
Admission	67.6±3.5	65.2±3.3	64.4±3.0	22.9±1.2	22.1±1.1	25.4±2.6	13.0±0.2	13.5±0.2	13.0±0.2
3 months	51.6±2.4***	52.8±3.3***	63.5±3.6	20.2±1.0*	21.8±1.2	24.1±1.9*	12.9±0.2	13.5±0.2	13.0±0.2
12 months	47.9±2.4***	47.8±2.8***	56.3±3.5**	22.0±2.6	19.7±0.8*	22.7±1.7**	13.1±0.2	13.5±0.2	12.8±0.2

	Hematocrit (%)		
Admission	40.4±0.5	41.6±0.6	40.4±0.5
3 months	40.4±0.5	41.7±0.6	40.1±0.4
12 months	41.0±0.5	42.6±0.6	40.0±0.6

* $p < 0.05$ vs corresponding admission value; ** $p < 0.01$ vs corresponding admission value; *** $p < 0.001$ vs corresponding admission value.

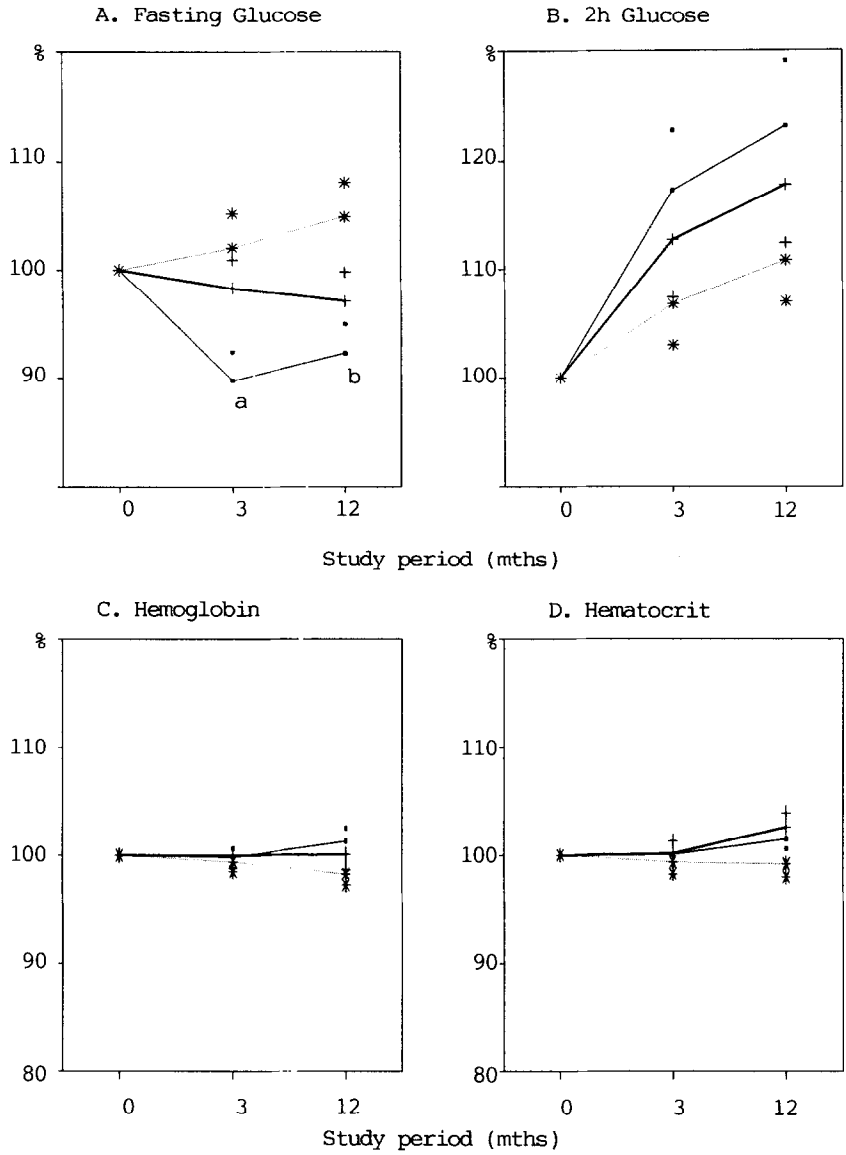

Fig. 1. Glucose, hemoglobin and hematocrit values (mean % of admission (0 mths) ± SEM) in Singaporean women on low-dose oral contraceptives and IUD. —•— NET/EE; —+— LNG/EE; —*— IUD; corresponding points above line represent mean ± SEM value; a: p <0.05 NET/EE vs LNG/EE or IUD; b: p <0.05 NET/EE vs IUD.

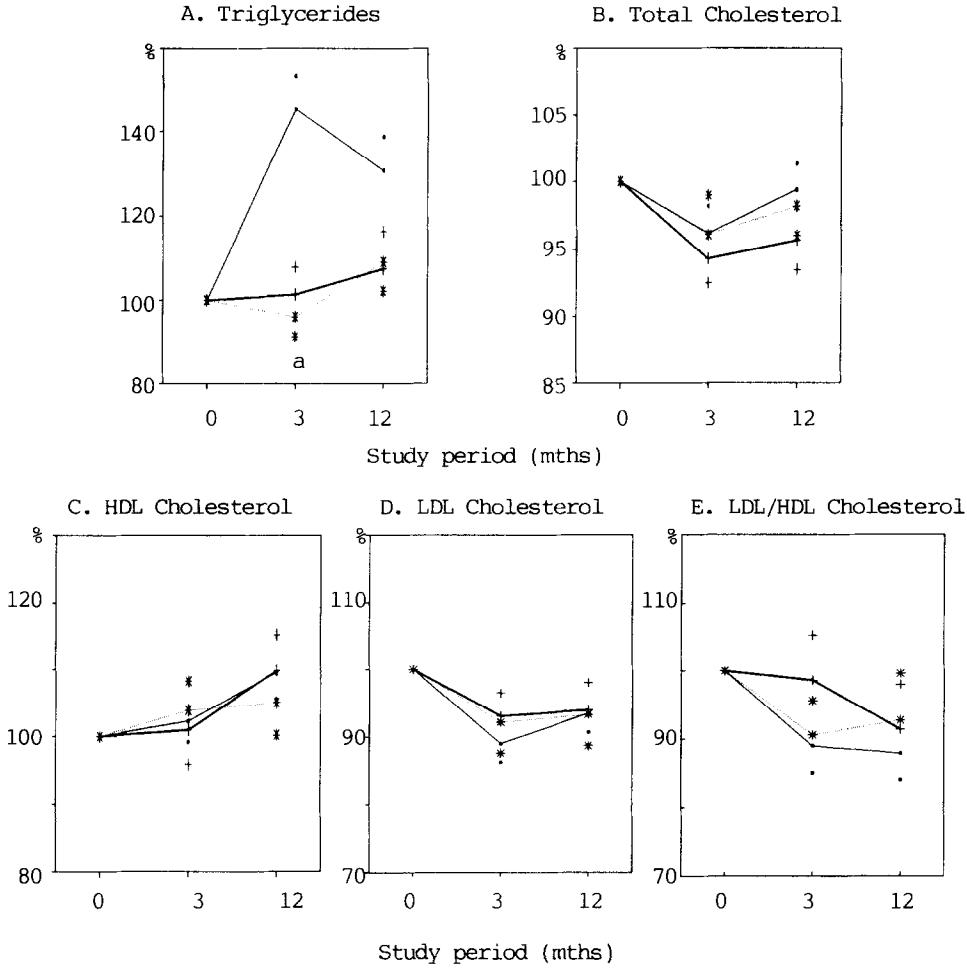

Fig. 2. Lipid values (mean % of admission (0 mths) ± SEM) in Singaporean women on low-dose oral contraceptives and IUD. —•— NET/EE; —+— LNG/EE; —*— IUD; corresponding points above line represent mean ± SEM value; a: p <0.05 NET/EE vs LNG/EE or IUD.

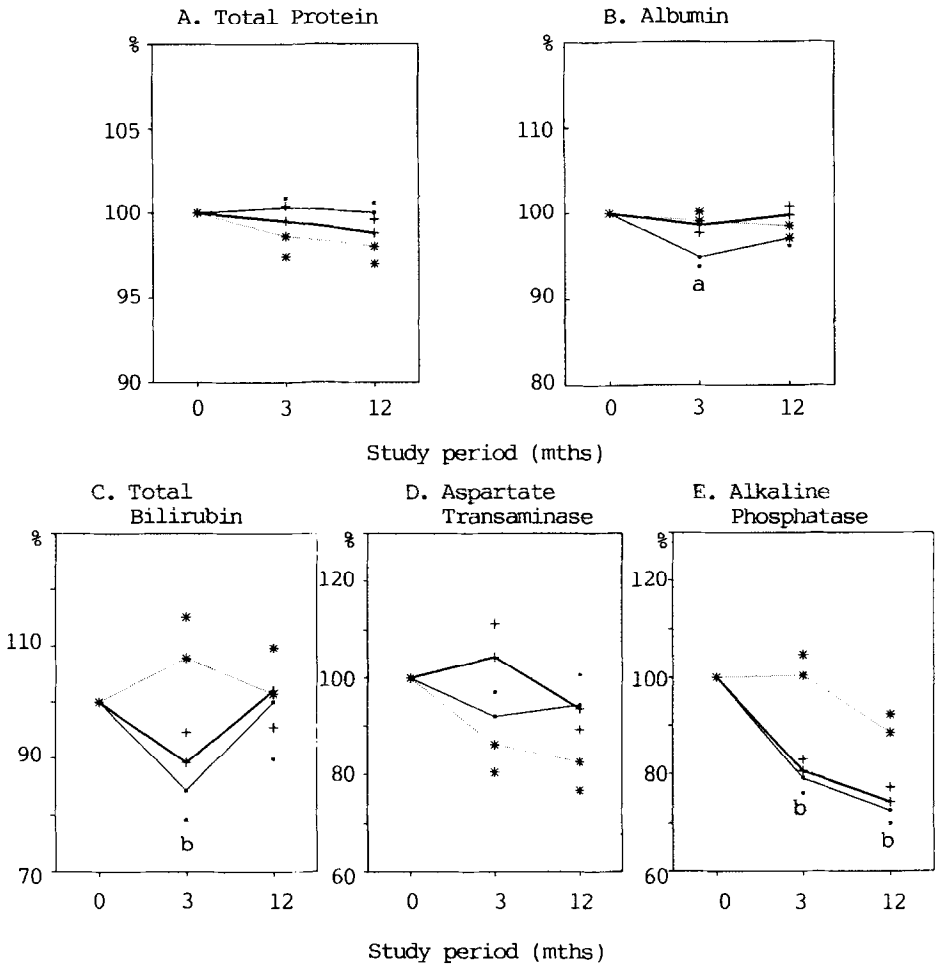

Fig.3. Total protein, albumin, total bilirubin, aspartate transaminase and alkaline phosphatase values (mean % of admission (0 mths) ± SEM) in Singaporean women on low-dose oral contraceptives and IUD. —•— NET/EE; —+— LNG/EE; —*— IUD; corresponding points above line represent mean ± SEM value; a: p <0.05 NET/EE vs LNG/EE or IUD; b: p <0.05 IUD vs NET/EE or LNG/EE.

Table IV. Number of abnormal cases of various metabolic parameters in Singaporean women on low-dose oral contraceptives and IUD

	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
	2h Glucose (>120 mg%)			Triglyceride (>174 mg%)			Total Cholesterol (>250 mg%)			HDL Cholesterol (<35 mg%)			LDL Cholesterol (>187 mg%)		
Admission	2	1	1	3	3	1	2	4	1	4	7	3	3	1	2
3 months	10(1)	4	2(1)	4(3)	3(2)	1	1(1)	3(3)	1	3	7(3)	2(1)	1(1)	2(1)	1(1)
12 months	12(1)	5	3(1)	3(2)	2(1)	2	1(1)	3(3)	2(1)	2(2)	3(3)	3(1)	1(1)	3(1)	2(1)
	Total Protein (>8.0 g/dl)			Alkaline Phosphatase (>95 U/l)			Total Bilirubin (>1.5 g/dl)			Aspartate Transaminase (>40 U/l)					
Admission	7	7	3	3	3	1	0	1	0	0	0	3			
3 months	8(6)	2(2)	4(2)	0	1(1)	1	1	1(1)	0	0	0	1			
12 months	9(6)	3(2)	3(2)	0	0	0	0	0	1	0	0	1			

N.B. Criteria of abnormality = limit of laboratory's reference range; numbers in brackets refer to number of cases with abnormal value on admission; no abnormal cases were observed for fasting glucose and albumin.

Abnormal parameters

Table IV shows the number of abnormal cases for each parameter. The normal ranges are based on the laboratory reference values (given in brackets after the parameter). In most of the parameters, abnormal cases were observed at admission in all groups, the highest number being in total protein. The greatest number of abnormal cases were observed for 2h glucose for NET/EE (12 cases including a case with continuously high values since admission) and LNG/EE (5 cases) at 12 months. For total protein, the number of abnormal cases after treatment remained high especially for NET/EE group mainly because many of these cases started off with high values at admission (6 cases were continuously high throughout the study).

For HDL cholesterol, both pill groups had a high number of abnormal cases at admission (4 and 7 for NET/EE and LNG/EE, respectively) but by 12 months, the number was reduced (2 and 3 for NET/EE and LNG/EE, respectively). In contrast, there was no change in the number of abnormals observed for IUD group. The number of abnormal LDL cases increased from 1 (at admission) to 3 (at 12 months) for LNG/EE group while it was the reverse for NET/EE group and remained constant for the IUD group. No abnormal values were observed for albumin for all groups. While 3 abnormal values were observed for alkaline phosphatase at admission for each of the pill group and 1 for the IUD group, all values returned to normal at 12 months. On admission, 1 case of abnormal bilirubin was observed in the LNG/EE group. At 3 months, the same case remained high but eventually returned to normal by 12 months. One abnormal value was observed at 3 months for the NET/EE group and 1 at 12 months for the IUD group. There was no abnormal value for SGOT in the pill groups on admission but 3 were observed in the IUD group. At 12 months, only 1 abnormal value remained in the IUD group while 1 abnormal value appeared in the NET/EE group.

Discussion

While changes in carbohydrate metabolism were observed with use of both pill preparations, these changes, especially the increases in 2h glucose, were not significantly different between pill groups and IUD. This may be due to a parallel significant increase in 2h glucose response in the IUD group at 12 months. The reason for the increasing trend in 2h glucose response in this control group is not known. These changes could not possibly be due to seasonal variation or laboratory factors since there are no extremes of temperature changes while laboratory performance has been strictly controlled and monitored by quality control programmes. Fasting glucose levels were, however, significantly lower in the NET/EE group than levels in IUD group at 3 and 12 months and LNG/EE group at 3 months only. In addition to decreased fasting glucose, Wynn et al. (3) also observed elevation of glucose tolerance curve in oral contraceptive preparations containing estrane or gonane progestagens. Similar to our findings with the pill preparation of LNG/EE, minimal

changes were observed in glucose tolerance in other studies (1,4,5). The difference in the varying response may be due to the composition of the pill. The diabetogenic effect has been attributed to the progestagen content. Estrogen by itself does not alter glucose tolerance (17), but in combination serves to inhibit the catabolism of progestagen (18).

NET/EE increased triglyceride and decreased LDL cholesterol levels throughout the treatment period. In contrast, LNG/EE hardly affected triglyceride levels while LDL cholesterol was significantly suppressed at 3 months, returning to normal by 12 months. Total cholesterol was also significantly suppressed by LNG/EE at both 3 and 12 months. While a similar trend was observed for total and LDL cholesterol in the IUD group, these changes were, however, not significantly different from their admission level. The varying response of each group to the different lipid parameters could be due to the interaction of both estrogen and progestagen component. Estrogen may increase triglycerides and HDL cholesterol and decrease LDL cholesterol, but progestagens with androgenic activity have the opposite effect (3,19,20). LNG being more androgenic than NET would exhibit more anti-estrogenic effect. While observed differences were seen within each pill group, it is important to note that except for triglycerides, these changes were not apparent when compared with the IUD control group. Furthermore, except for the increased triglycerides (which may add to cardiovascular risk if associated with a fall in HDL cholesterol, though this is not clearly defined), all changes in lipids seemed favorable to a decreased cardiovascular risk (lowering of atherogenic index).

Van der Vange et al. did not observe any significant differences in total and HDL cholesterols as well as triglycerides between women on NET/EE and LNG/EE for 6 months (21). Roy et al. observed slight decrease in total cholesterol and no change in triglycerides in women on LNG/EE for 3 and 6 months (22). Larsson-Cohn et al. observed a significant decrease in total cholesterol at 1 month, returning to normal by 3 and 6 months while triglycerides was significantly increased at 3 months and HDL cholesterol remained low throughout the study period (23). In another study in our department with a parallel control group, comparing the lipoprotein effects of a monophasic and triphasic preparation of LNG/EE, similar results were obtained for triglycerides, total, HDL and LDL cholesterols in the monophasic LNG/EE with the same dosage (24). However, HDL cholesterol was significantly lower at 3 and 12 months and total cholesterol at 12 months than the IUD group. In the current study, there was no significance observed between the LNG/EE and IUD groups. This difference was probably due to the smaller sample size in the previous study (15 compared to 29).

Both NET/EE and LNG/EE did not affect hemoglobin, hematocrit and total protein values but suppressed alkaline phosphatase up to 12 months and total bilirubin up to 3 months. While albumin was significantly suppressed up to 12 months and SGOT was significantly suppressed initialy at 3 months in NET/EE users, albumin was unchanged and SGOT significantly suppressed later at

12 months in LNG/EE users. In the IUD (control) group, alkaline phosphatase was significantly suppressed at 3 months and SGOT at 3 and 12 months. From mean levels of SGOT at admission, it may seem that the IUD group had higher mean levels than the pill groups. The reason for this is not known. Ahren et al. (4) and Roy et al. (22) observed similar results for hemoglobin, hematocrit, total protein, albumin and alkaline phosphatase in LNG/EE users. Ahren et al. (4) observed a similar decrease in total bilirubin at 3 months, returning to normal at 12 months but SGOT was unchanged throughout the study period, while Roy et al. (22) did not observe any changes in either total bilirubin or SGOT.

While differences were obtained in individual treatment groups, the differences in liver function were seen between different treatment groups only for albumin which was lower in NET/EE than LNG/EE or IUD groups at 3 months only and both pill groups had lower alkaline phosphatase than the IUD group. Because of the parallel variation in SGOT levels in the IUD group, differences between pill and IUD groups were not observed. The earlier pills consisting of very high doses of estrogen were associated with increases in SGOT (25-27). With reduction in the dosage of estrogen, there were no significant elevations or even lower values were obtained (4,22,28,29).

Except for 2h glucose, the number of abnormal values for other parameters did not increase substantially after treatment. For most parameters, there was a reduction of the number of abnormal values by 12 months. The number of abnormal 2h glucose was increased in all groups (10/29 or 34.5 %, 4/29 or 13.8 % and 2/23 or 8.7 % for NET/EE, LNG/EE and IUD users, respectively, by 12 months). Thus, except for 2h glucose, the effects of the pills may be considered as minimal or even an improvement on other parameters. Furthermore, the observed effects in most metabolic parameters were reduction in values which may not have any clinical significance.

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